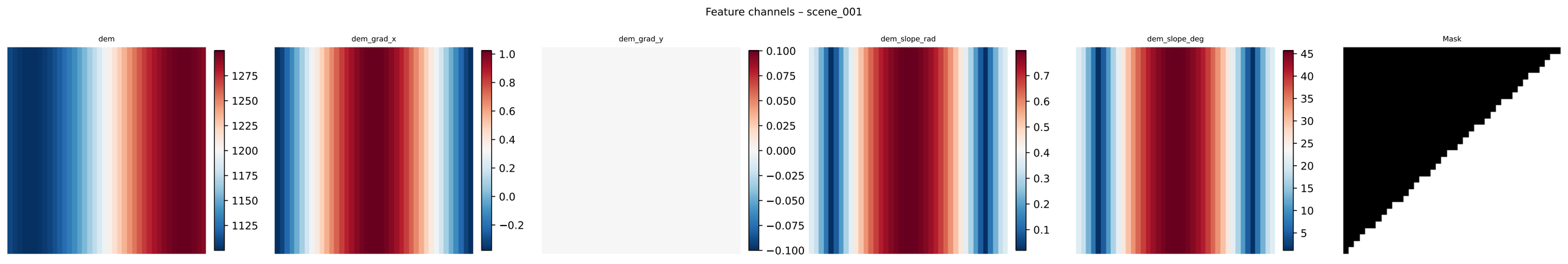
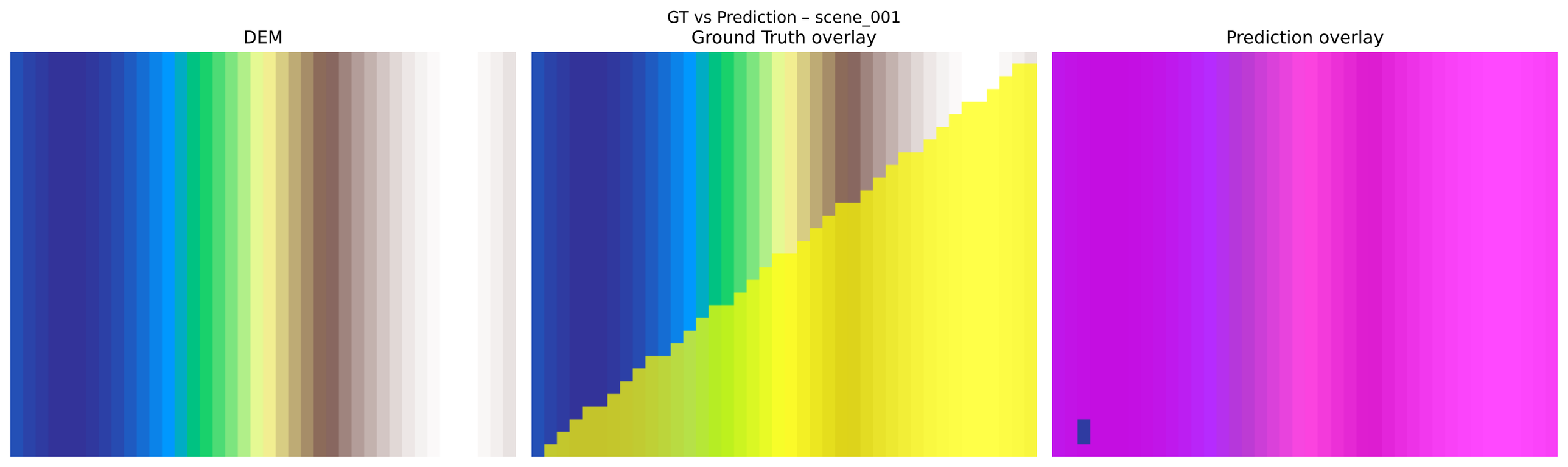
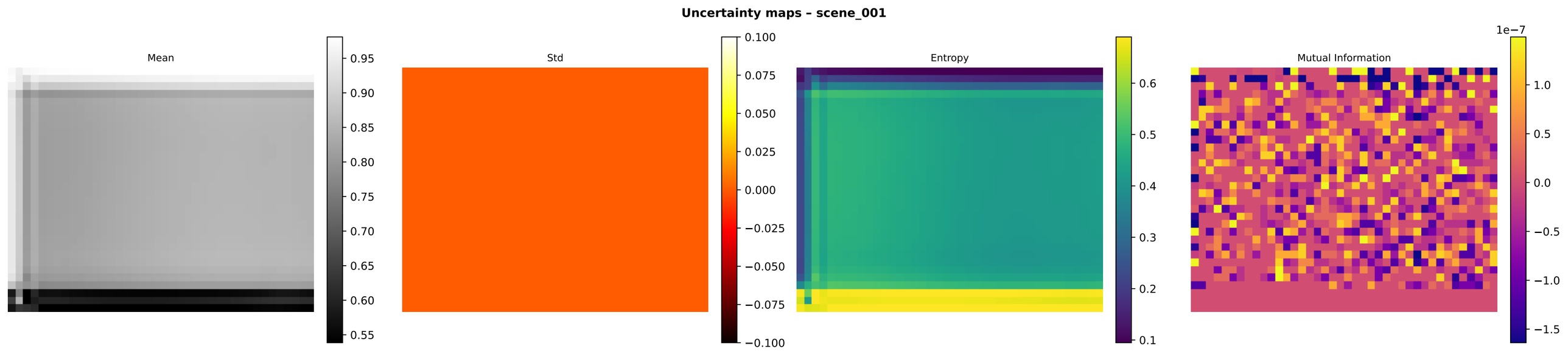
Multisource Inference — Uncertainty Maps Report

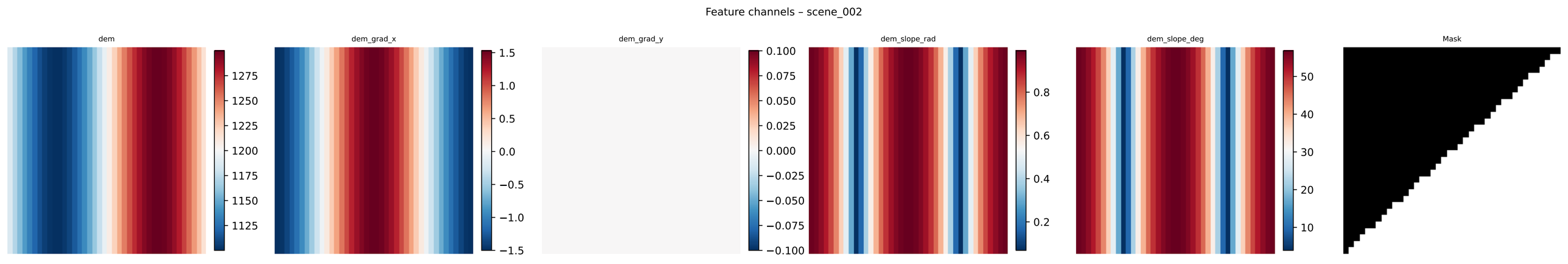
MC-Dropout uncertainty on scene_001 and scene_002

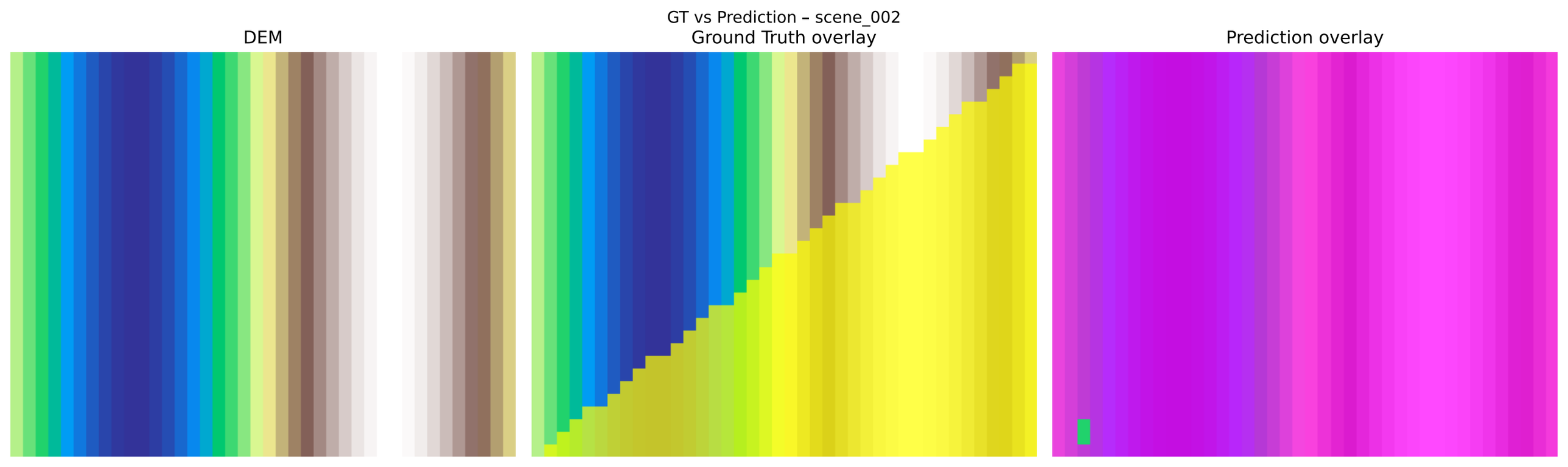
Model: unet26_ckpt.pt
Channels: 26
MC samples: 30
Scenes: scene_001, scene_002

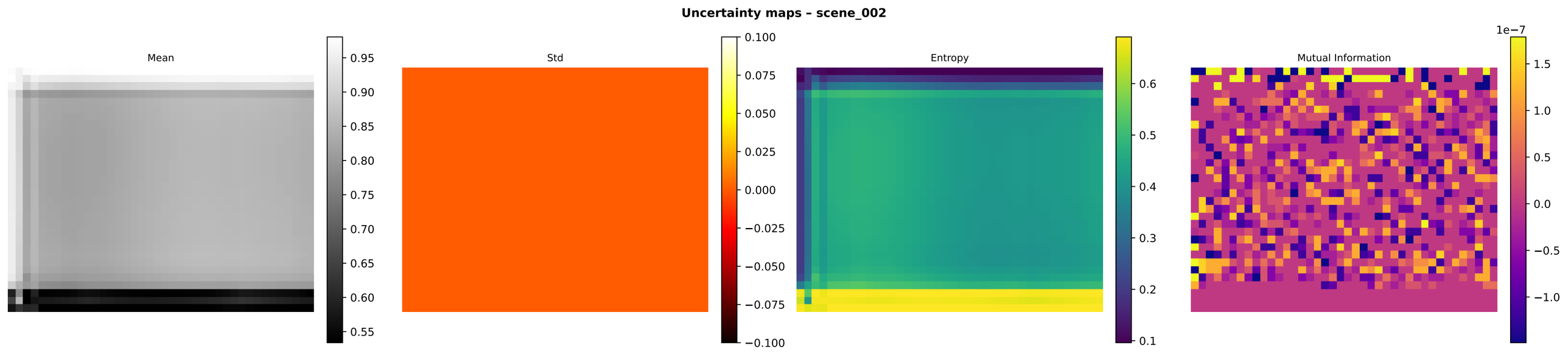

Feature channels – scene_001
dem
1275
1250
1225
1200
1175
1150
1125
dem_grad_x
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
dem_grad_y
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
dem_slope_rad
0.7
0.6
0.5
0.4
0.3
0.2
0.1
dem_slope_deg
45
40
35
30
25
20
15
10
5
Mask


GT vs Prediction – scene_001
DEM
Ground Truth overlay
Prediction overlay


Uncertainty maps – scene_001
Mean
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
Std
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
Entropy
0.6
0.5
0.4
0.3
0.2
0.1
Mutual Information
1e−7
1.0
0.5
0.0
−0.5
−1.0
−1.5


Feature channels – scene_002
dem
1275
1250
1225
1200
1175
1150
1125
dem_grad_x
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
dem_grad_y
dem_slope_rad
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
0.8
0.6
0.4
0.2
dem_slope_deg
50
40
30
20
10
Mask


GT vs Prediction – scene_002
DEM
Ground Truth overlay
Prediction overlay


Uncertainty maps – scene_002
Mean
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
Std
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
Entropy
0.6
0.5
0.4
0.3
0.2
0.1
Mutual Information
1e−7
1.5
1.0
0.5
0.0
−0.5
−1.0